Req3 shared plots for comparison

Branch: feature/req3-primal-dual

Shared input folder: results/processed/req4_shared_realizations

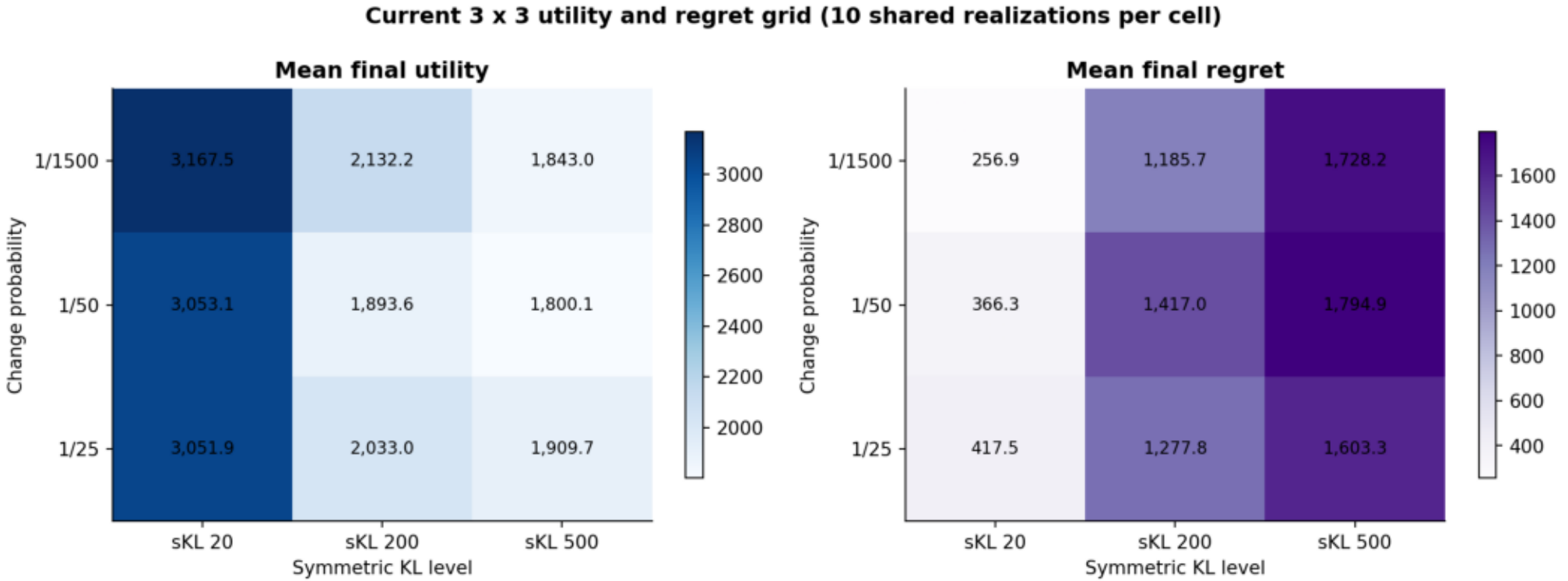
Settings: N=5, B=2500, sKL={20, 200, 500}, p={1/1500, 1/50, 1/25}

Low-frequency schedule: 2 change points at rounds 802 and 1988, minimum gap 802 rounds.

Conflict graph: 5-cycle; feasible super-arms are 5 singletons + 5 non-adjacent pairs.

Full 3 x 3 grid

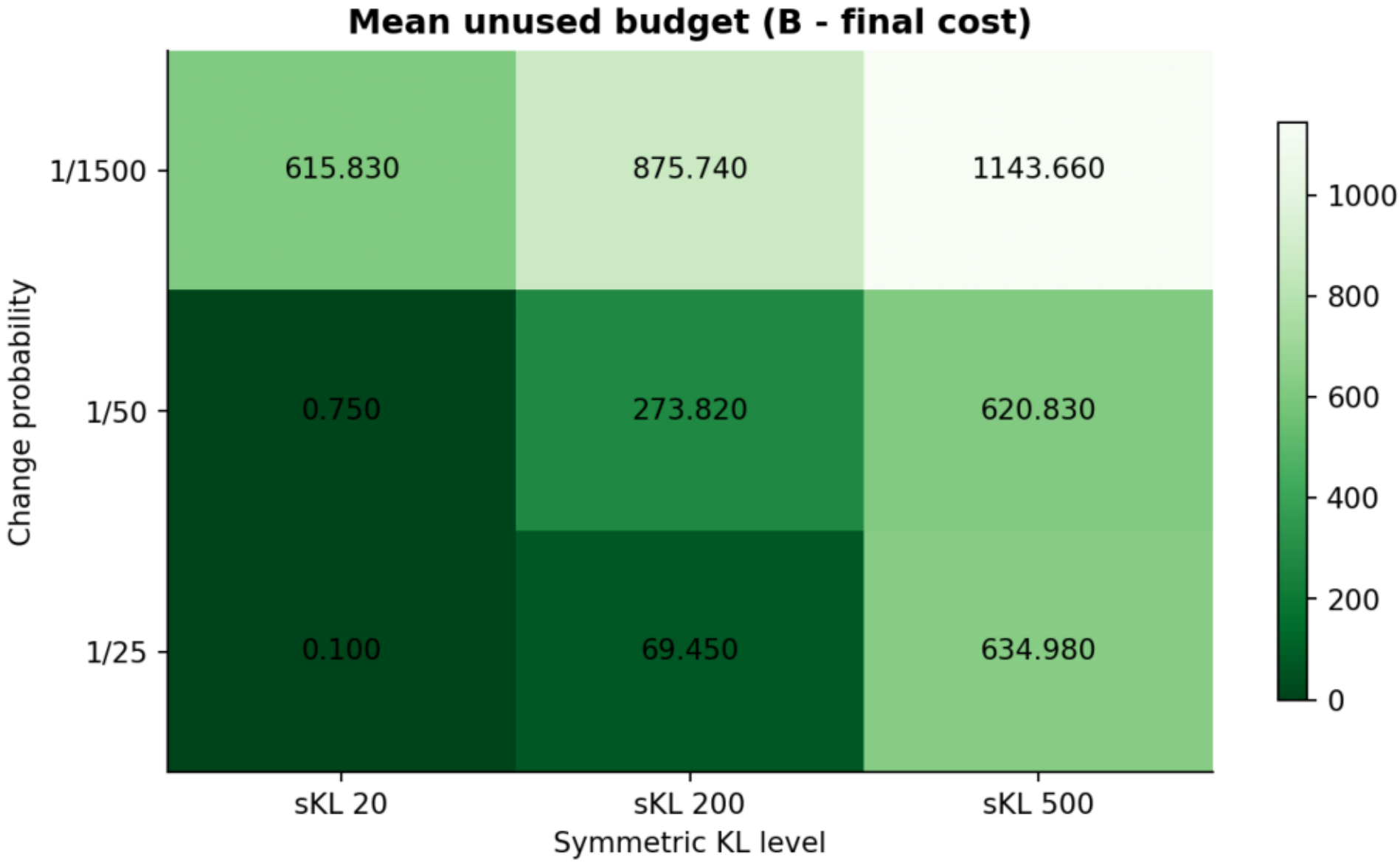
Mean final utility and regret over 10 shared realizations per cell. This is the quickest overview of how sKL and change frequency affect Req3.



sensitivity_grid.png

Budget feasibility

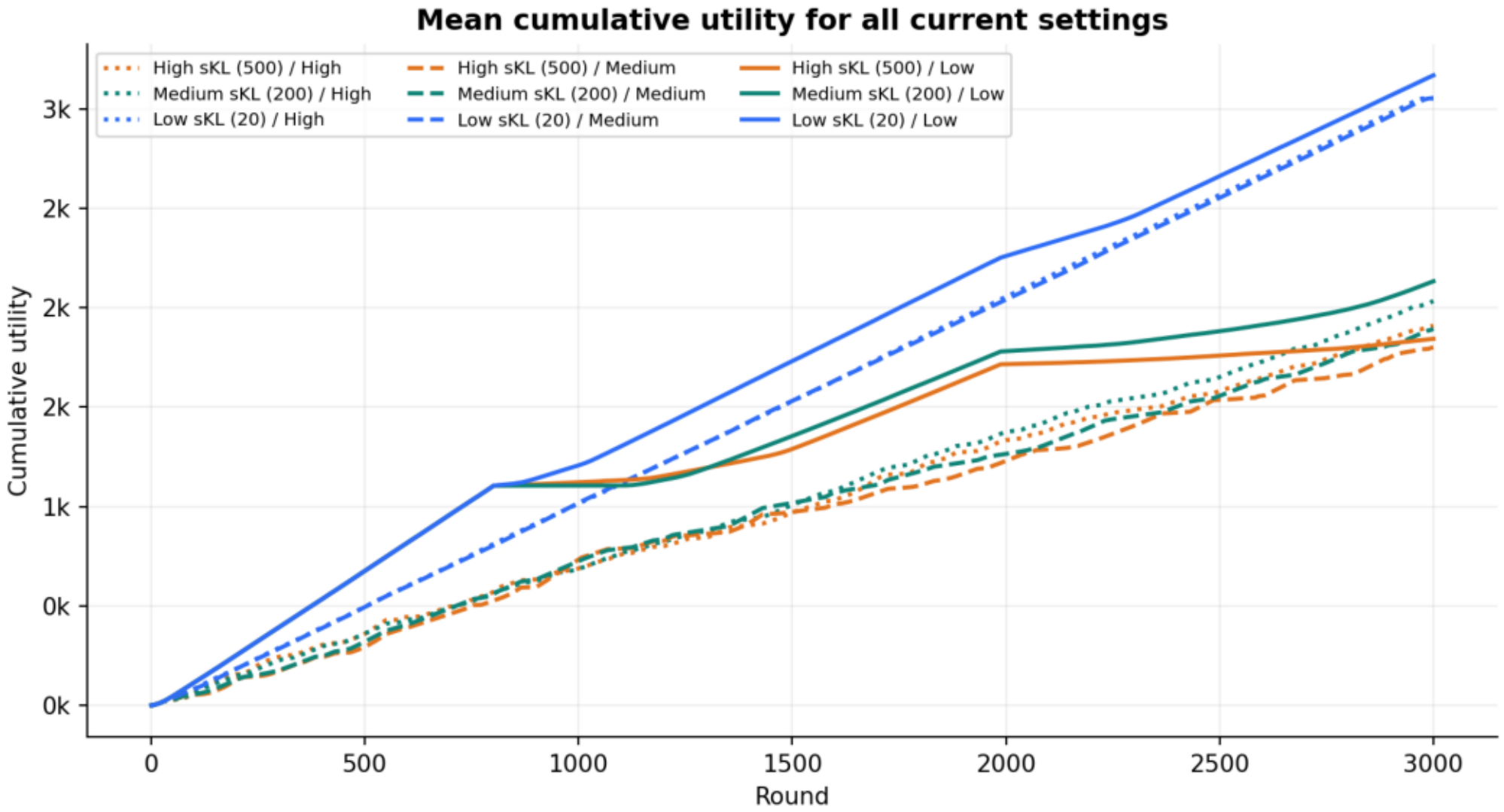
Unused budget B - final cost with B=2500. All runs satisfy the hard budget constraint with 0 violations.



budget_feasibility.png

Cumulative utility curves

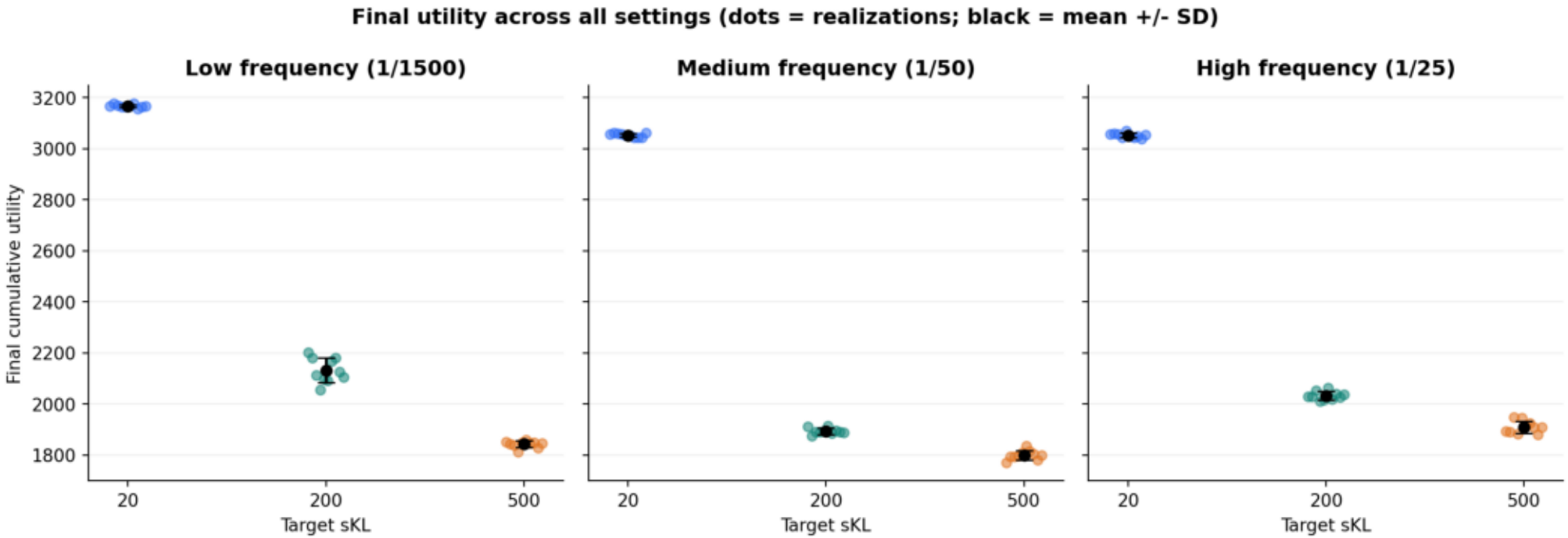
Mean learner cumulative utility for all 9 settings. sKL has the strongest visible effect; frequency is secondary.



all_learning_curves.png

Final utility distribution

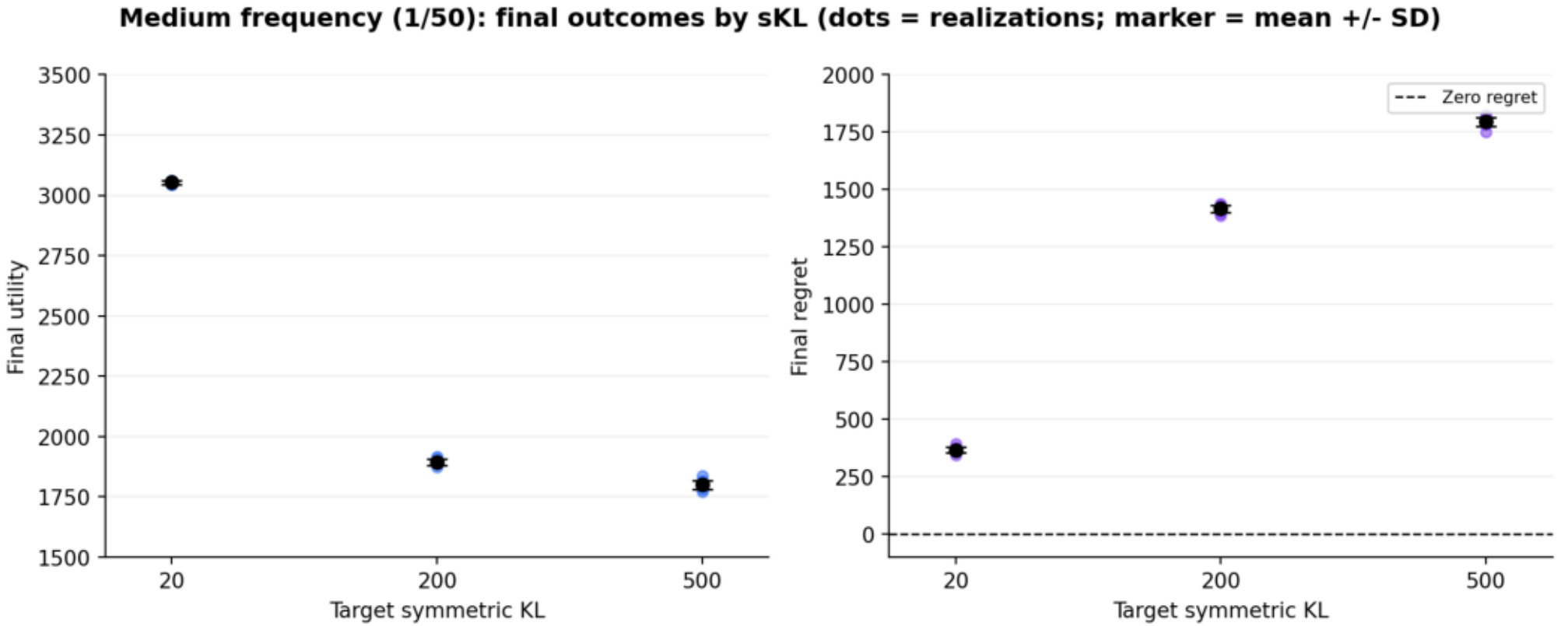
Each dot is one realization; black markers show mean +/- SD. This shows variability across the 10 shared realizations.



all_setting_final_utility_distribution.png

Effect of sKL at medium frequency

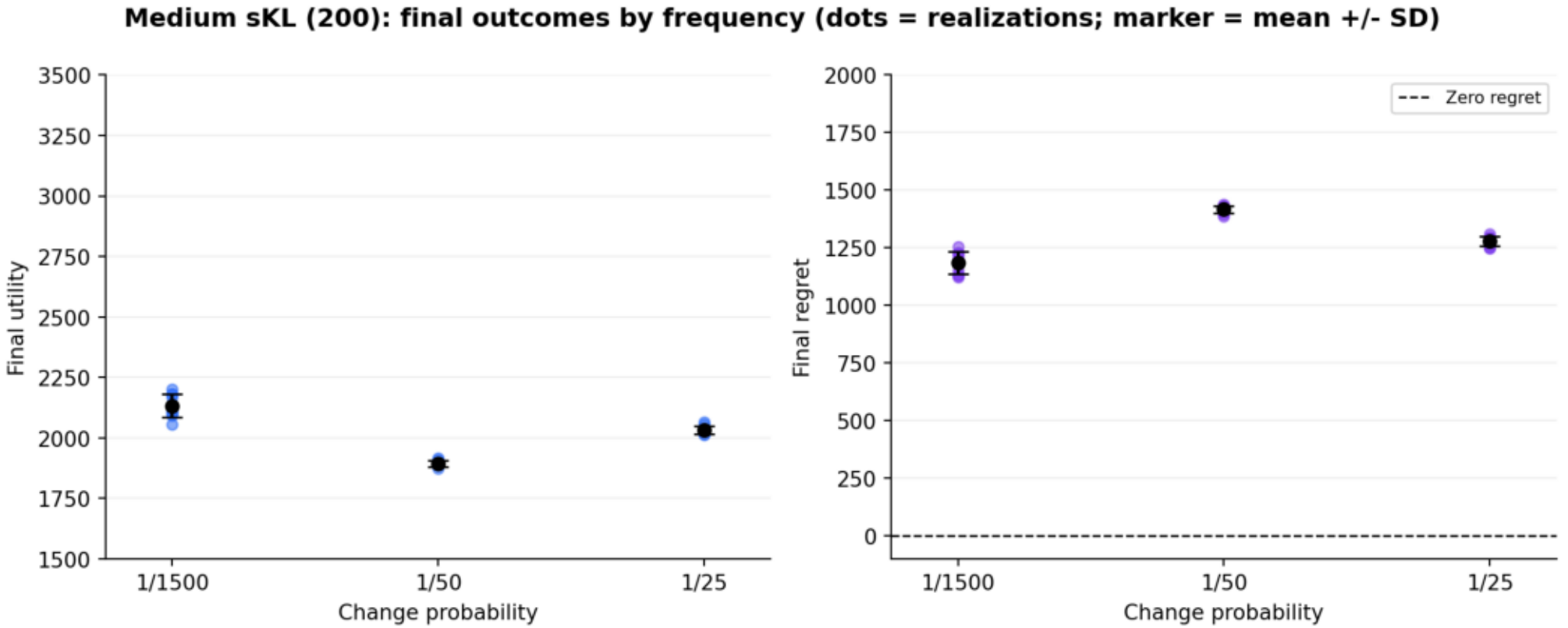
Final utility and regret when p=1/50. This isolates the effect of Gaussian separation.



summary_skl.png

Effect of frequency at medium sKL

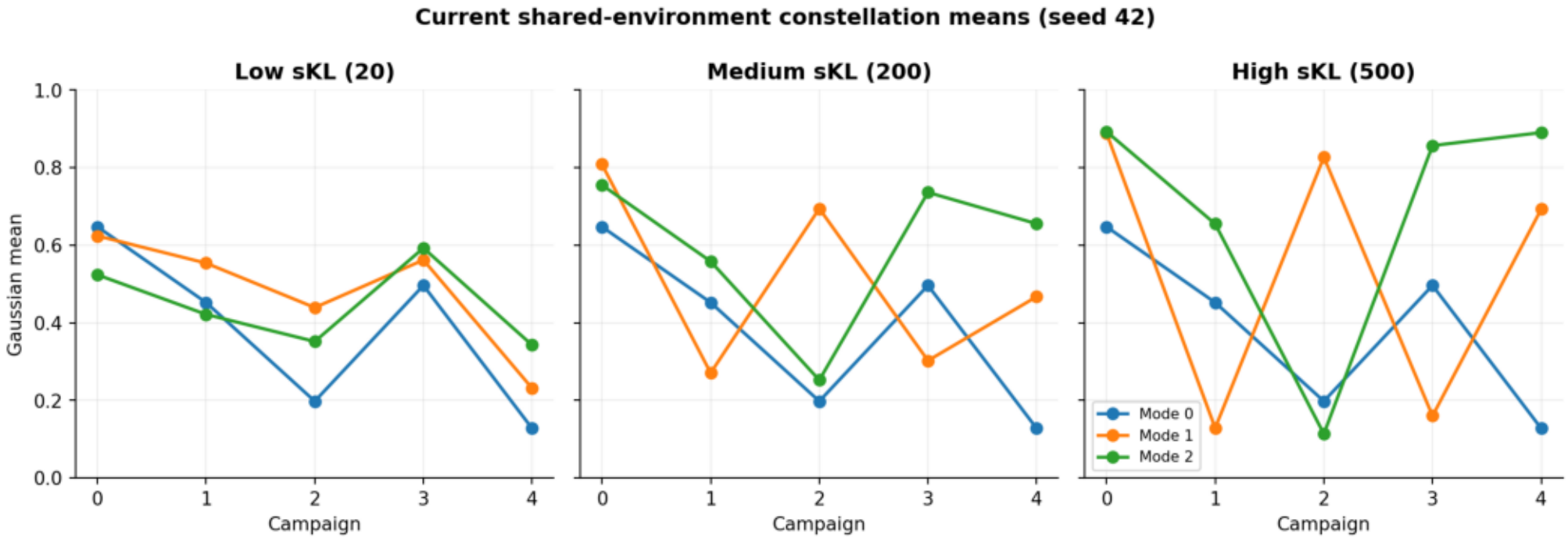
Final utility and regret when sKL=200. Low frequency is now p=1/1500 with only 2 well-separated changes.



summary_frequency.png

Constellation means

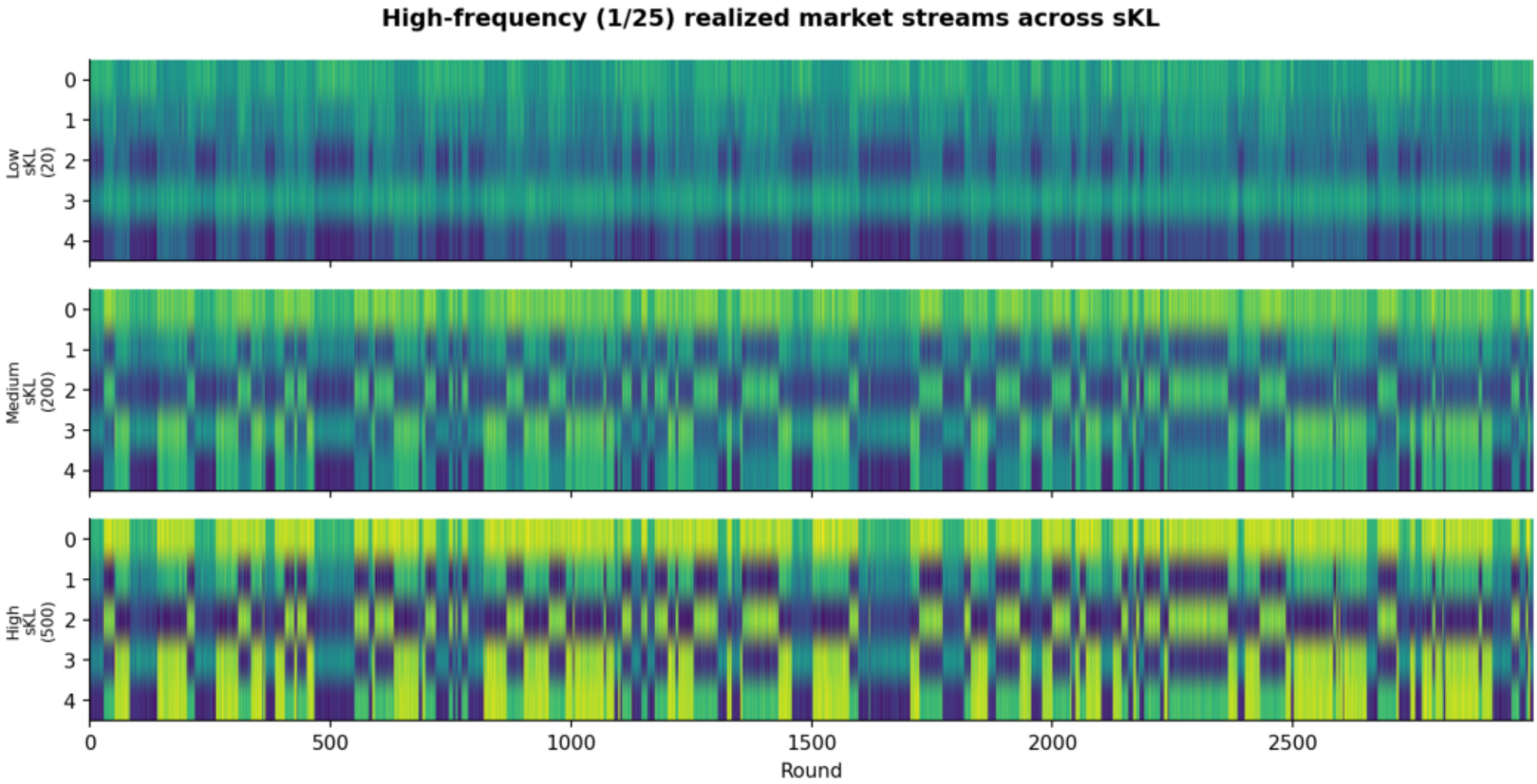
Gaussian mode means saved in constellation.npz for the three sKL levels.



constellation_profiles.png

Example realized market streams

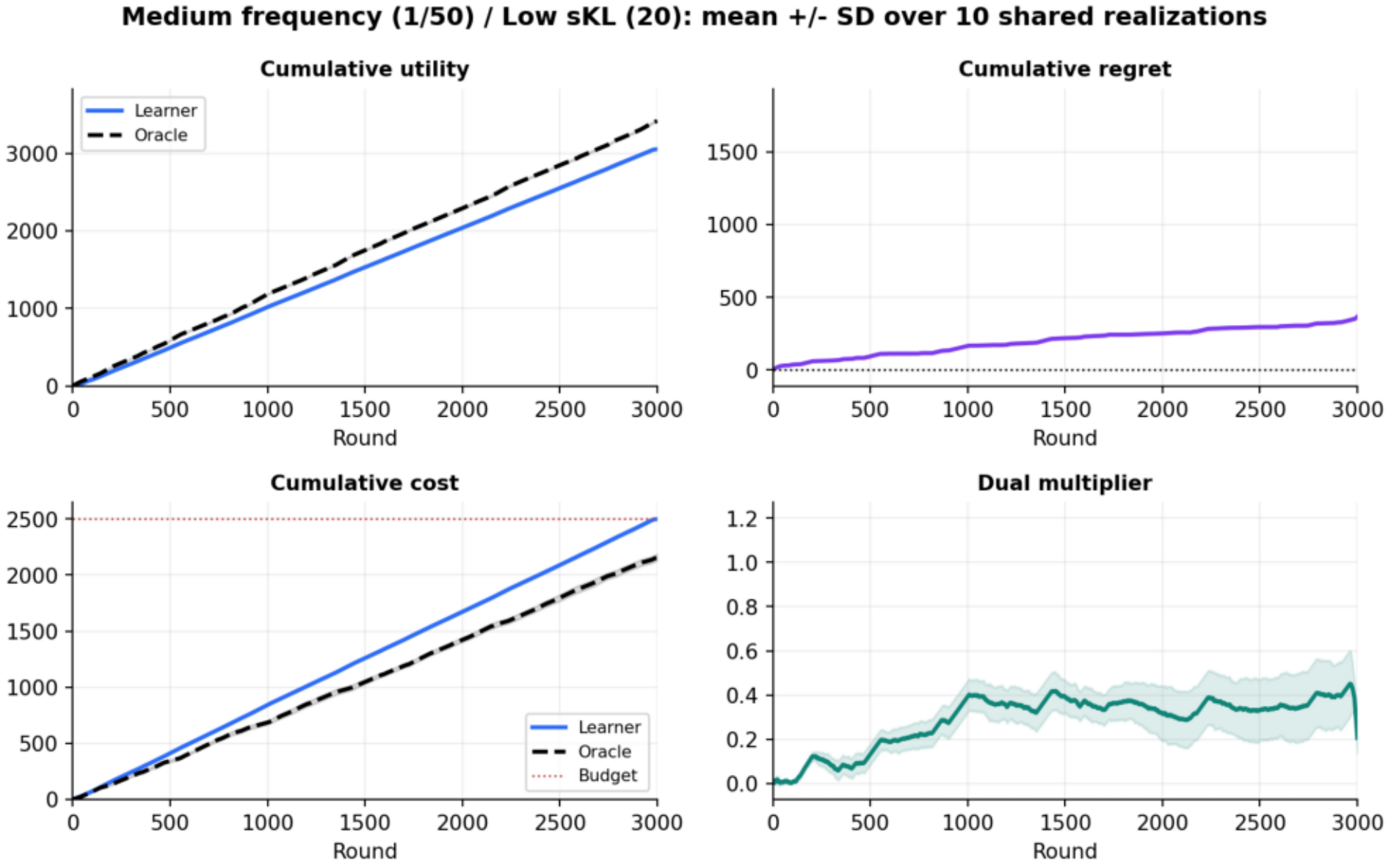
High-frequency p=1/25 streams for sKL 20, 200, and 500. This visualizes the fixed market inputs.



high_frequency_spectrum.png

Low sKL diagnostic at medium frequency

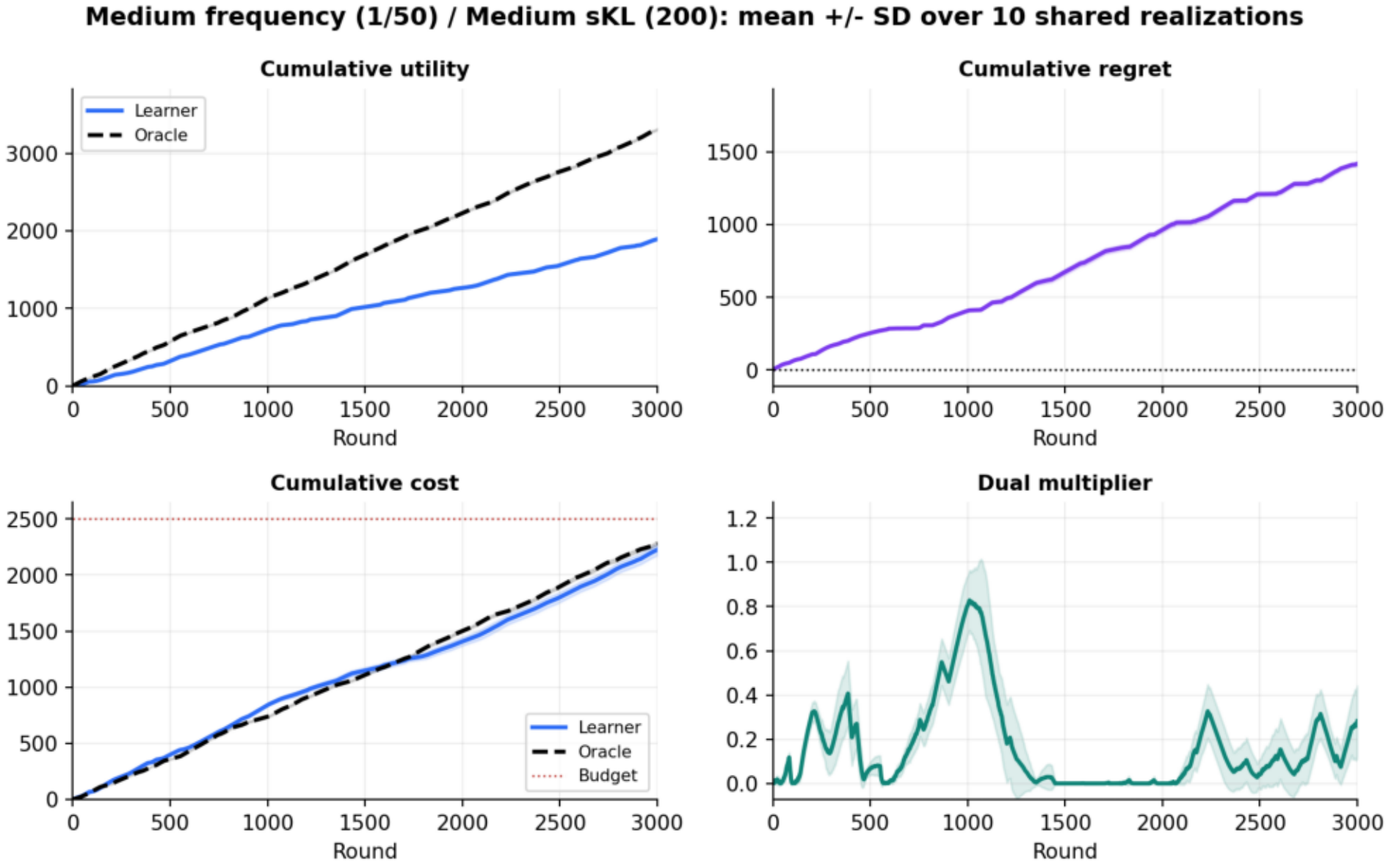
Detailed dynamics for sKL=20, p=1/50: utility, regret, cost, and dual multiplier.



diagnostic_medium_frequency_low_skl.png

Medium sKL diagnostic at medium frequency

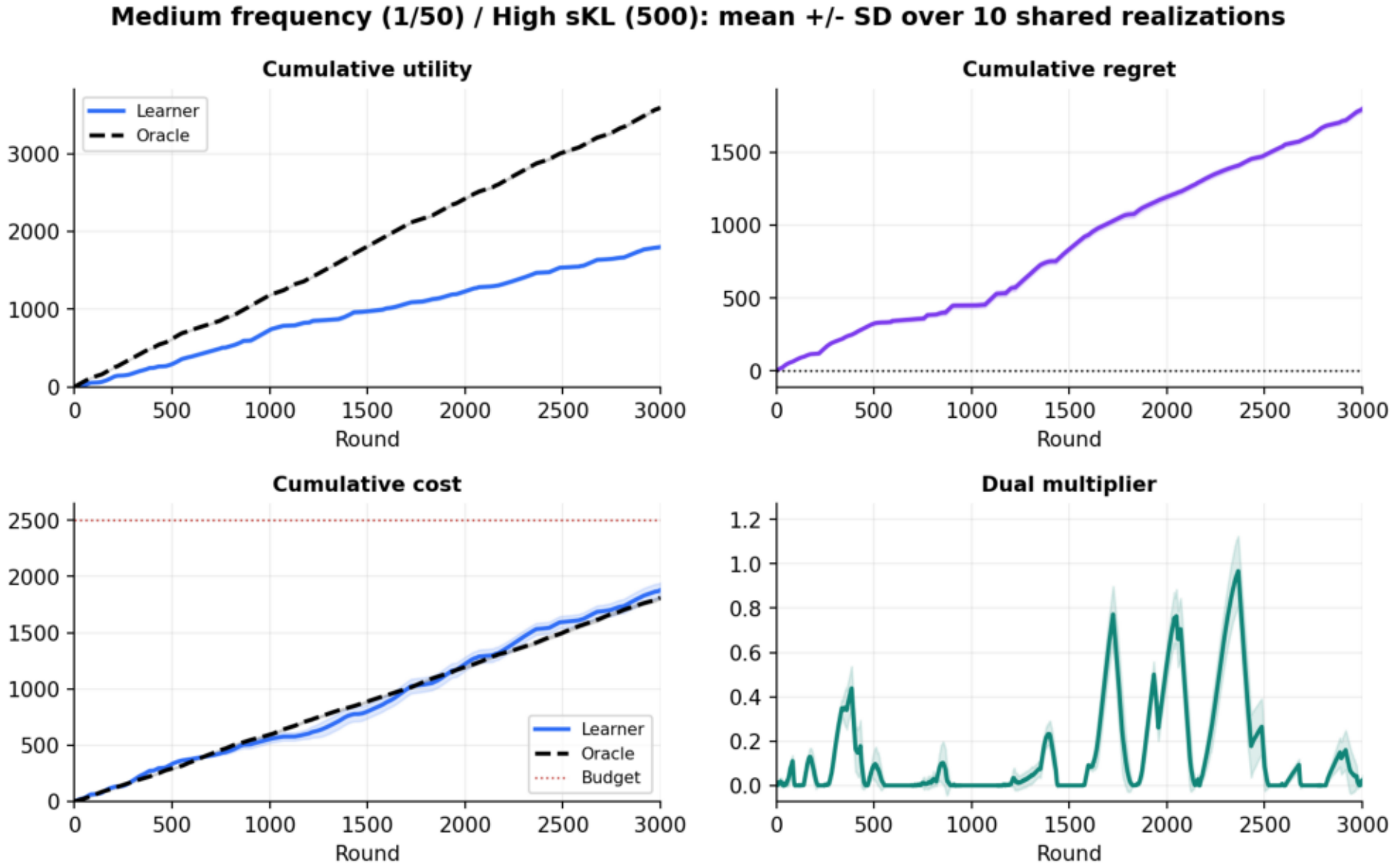
Representative current run dynamics for sKL=200, p=1/50.



diagnostic_medium_frequency_medium_skl.png

High sKL diagnostic at medium frequency

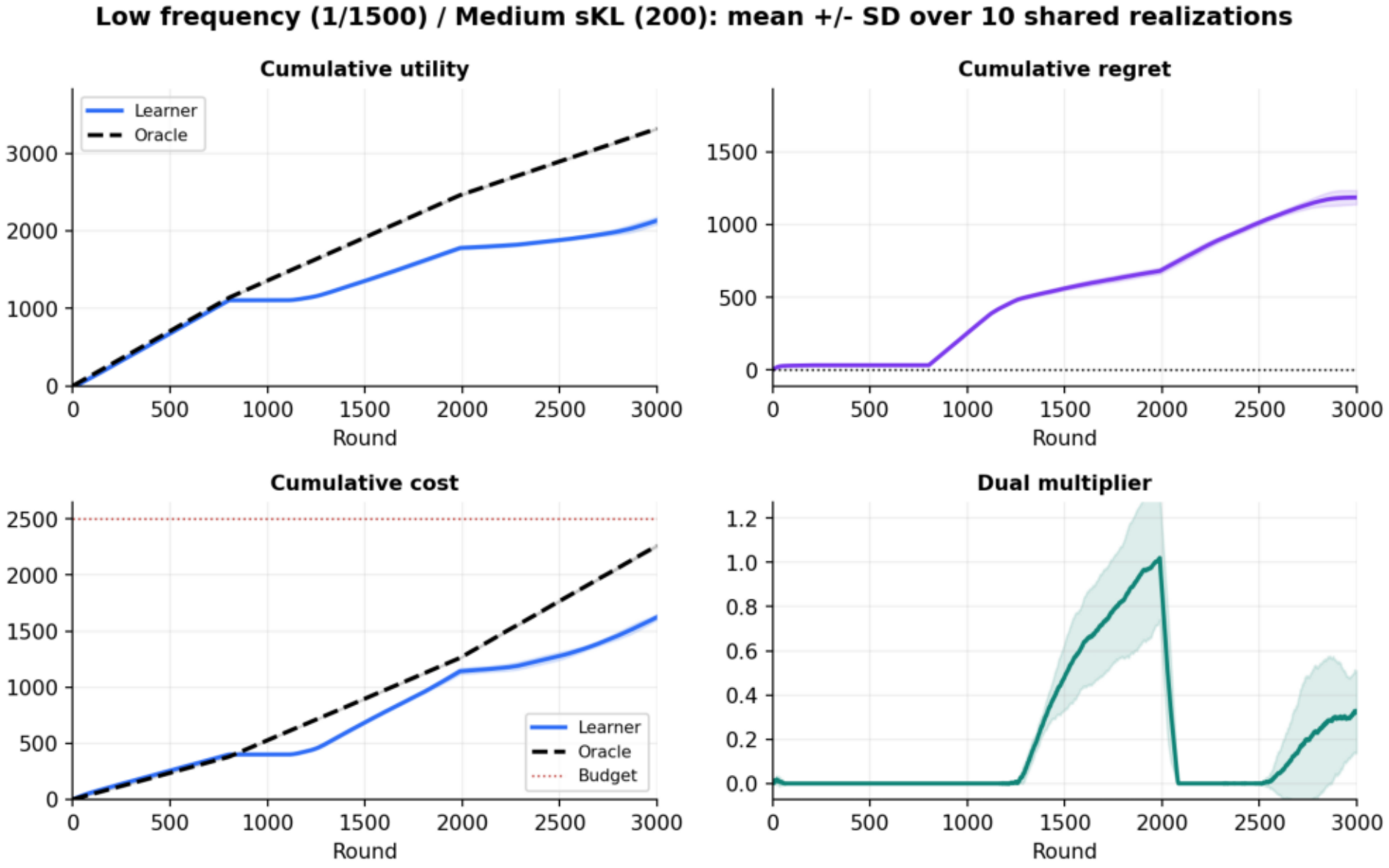
Detailed dynamics for sKL=500, p=1/50.



diagnostic_medium_frequency_high_skl.png

Low-frequency diagnostic at medium sKL

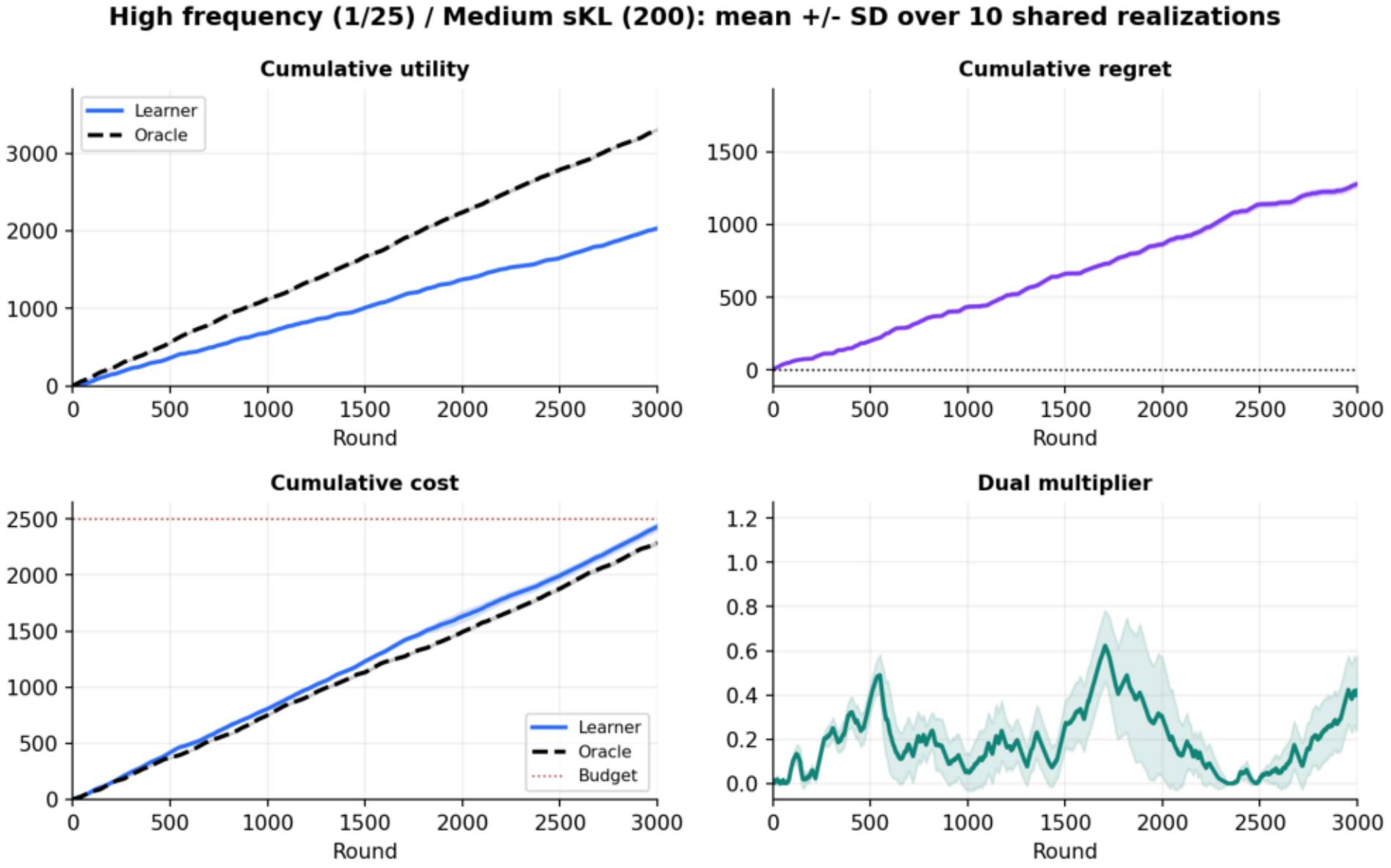
Detailed dynamics for p=1/1500, sKL=200.



diagnostic_low_frequency_medium_skl.png

High-frequency diagnostic at medium sKL

Detailed dynamics for p=1/25, sKL=200.



diagnostic_high_frequency_medium_skl.png